

Projector-Mass First-Alias Threshold and an Isospectral Cell Obstruction

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August 2026

Abstract

The projector-density gauge of RH-335 turns the global noisy parity scalar into signed localized cells. We determine the exact first-alias scale of one such cell. On every bounded-phase natural clock, its parity contribution divided by $H_k = kR^{-2k}$ equals $2C_*\lambda^{\eta_\sigma}(\beta R)^{2k}\pi_\sigma(J)(1+o(1))$. Thus the critical projector-mass exponent is $\kappa_{\text{proj}} = \log(\beta R)/\log \lambda$, distinct from the earlier Duhamel stability exponent. A fixed finite partition necessarily has a maximum normalized parity cell diverging to $+\infty$, but the maximizing cell may move with noise and is not identified with a physical boundary/sibling aggregate; raw and alias terms may still cancel it. We complement the scale theorem with an exact three-state family. A nontrivial interval of smooth similarities preserves strict positivity, row-stochasticity, the spectrum, and every power trace, while the parity projector masses and corrected singleton cells drift by explicit nonzero zero-sum vectors. This is a nonphysical finite algebraic obstruction to recovering cells from global spectral data. It does not prove physical signed Duhamel cancellation or a physical normalized obstruction, and it yields no determinant or Riemann-hypothesis conclusion.

1 Inherited projector gauge and first-alias clock

Let K_σ be the noisy folded operator and let $\lambda_-(\sigma) = -(1 - \delta_\sigma)$ be its real simple parity eigenvalue, where the archived law is

$$\delta_\sigma = C_*\sqrt{\sigma} + o(\sqrt{\sigma}), \quad C_* > 0. \quad (1)$$

Write $E_{-, \sigma}$ for the rank-one Riesz projector itself. RH-335 proves that

$$\pi_\sigma(J) = \text{Tr}(M_J E_{-, \sigma}) \quad (2)$$

is a real finite signed measure with total mass one [3]. Allocating the deterministic parity scalar with this noisy density is a frozen gauge, not canonical physical localization or deterministic/noisy projector transport.

Retain the constants and clock from RH-326 [1]:

$$r_H = \frac{17}{20}, \quad R = \frac{7}{5}, \quad \beta = \frac{1}{r_H \sqrt{\lambda}}, \quad (3)$$

and

$$\eta_\sigma = k - \frac{\log(1/\sigma)}{2 \log \lambda}. \quad (4)$$

Throughout the moving statements, η_σ remains bounded and $k \rightarrow \infty$ as $\sigma \rightarrow 0$.

For a measurable cell J , possibly depending on σ , define

$$\mathcal{G}_{\sigma, k}(J) = r_H^{-2k} \{1 - \lambda_-(\sigma)^{2k}\} \pi_\sigma(J), \quad H_k = kR^{-2k}. \quad (5)$$

This is only the localized parity-gauge constituent of the RH-335 corrected cell. It omits the localized raw defect and the globally subtracted alias packet.

2 Moving projector-mass threshold

Theorem 2.1 (Exact first-alias projector-mass scale). *On every bounded-phase sequence,*

$$\boxed{\frac{\mathcal{G}_{\sigma,k}(J)}{H_k} = 2C_* \lambda^{\eta_\sigma} (\beta R)^{2k} \pi_\sigma(J) \{1 + o(1)\}.} \quad (6)$$

The scalar $o(1)$ is independent of the chosen cell.

Proof. Because $k\delta_\sigma \rightarrow 0$, the uniform parity expansion of RH-326 gives

$$1 - (1 - \delta_\sigma)^{2k} = 2kC_* \sqrt{\sigma} \{1 + o(1)\}. \quad (7)$$

Even order removes the sign of λ_- , so substitution in (5) and division by H_k yield

$$2C_* \sqrt{\sigma} \left(\frac{R}{r_H} \right)^{2k} \pi_\sigma(J) \{1 + o(1)\}.$$

Equation (4) gives $\sqrt{\sigma} \lambda^k = \lambda^{\eta_\sigma}$, while $(\beta R)^{2k} = (R/r_H)^{2k} \lambda^{-k}$. Their product is the required scalar in (6). $\square$

Define

$$\boxed{\kappa_{\text{proj}} = \frac{\log(\beta R)}{\log \lambda} = \frac{\log(28/17)}{\log \lambda} - \frac{1}{2}.} \quad (8)$$

The rational bracket $3/2 < u_c < 25/16$ from the physical observation source implies $\lambda = 2u_c(u_c - 1) < 225/128 < (28/17)^2$ [2]. Hence $\beta R > 1$ and $\kappa_{\text{proj}} > 0$. Ordinary substitution of the archived decimal for λ gives

$$\kappa_{\text{proj}} = 0.463406944517002 \dots \quad (9)$$

This decimal is diagnostic, not an interval certificate. It is not the RH-325 sufficient Duhamel weight ceiling $\gamma_* = 0.3503698834605293 \dots$; the two exponents govern different objects.

Corollary 2.2 (Negligibility, exact phase conversion, and critical limit). *On every bounded-phase sequence,*

$$\mathcal{G}_{\sigma,k}(J) = o(H_k) \iff \pi_\sigma(J) = o((\beta R)^{-2k}). \quad (10)$$

Moreover,

$$\boxed{(\beta R)^{-2k} = \sigma^{\kappa_{\text{proj}}} (\beta R)^{-2\eta_\sigma}.} \quad (11)$$

If

$$(\beta R)^{2k} \pi_\sigma(J) \longrightarrow p, \quad \eta_\sigma \longrightarrow \eta, \quad (12)$$

then

$$\frac{\mathcal{G}_{\sigma,k}(J)}{H_k} \longrightarrow 2C_* \lambda^{\eta_\sigma} p. \quad (13)$$

Proof. Bounded phase makes $2C_* \lambda^{\eta_\sigma} \{1 + o(1)\}$ bounded above and away from zero. This proves (10) from theorem 2.1. From (4), $k = \log(1/\sigma)/(2 \log \lambda) + \eta_\sigma$; inserting this into the left side of (11) proves the exact identity. Finally apply (12) to (6). $\square$

3 A forced large parity cell, but not a physical obstruction

Theorem 3.1 (Fixed-partition maximum). *Let $\mathcal{P} = \{J_1, \dots, J_N\}$ be one fixed finite measurable partition. Then*

$$\max_{1 \leq i \leq N} \pi_\sigma(J_i) \geq \frac{1}{N}. \quad (14)$$

Consequently,

$$\max_{1 \leq i \leq N} \frac{\mathcal{G}_{\sigma,k}(J_i)}{H_k} \longrightarrow +\infty. \quad (15)$$

The maximizing index may depend on σ . Along every sequence $\sigma_\ell \rightarrow 0$, some fixed index recurs as a maximizer on a subsequence, and its normalized parity cell diverges on that subsequence.

Proof. RH-335 gives $\sum_{i=1}^N \pi_\sigma(J_i) = 1$, so the maximum is at least the arithmetic mean $1/N$, even though individual masses may be negative. The scalar in (6) is positive for small noise. Since bounded phase makes λ^{η_σ} uniformly bounded below and $\beta R > 1$, (14) proves (15). The subsequence statement is the pigeonhole principle on the finite index set. $\square$

Remark 3.2 (Cancellation firewall). The theorem concerns only the parity-gauge constituent. It does not identify a maximizing cell with the physical $\mathcal{B} + \mathcal{S}$ aggregate, and it does not prove a physical nonzero normalized obstruction. The localized raw defect can cancel the parity term inside a corrected cell, other cells can cancel in the partition sum, and the first-alias coefficient further subtracts $\mathcal{A}_{k,2k}$. The maximizing index may also move with noise. Thus physical signed $\Delta_B + \Delta_S$ cancellation remains NOT TESTABLE.

4 A strictly positive isospectral Markov family

We now give a separate finite algebraic witness. Reuse the exact RH-335 matrix and parity projector

$$K = \begin{pmatrix} 3/17 & 7/51 & 35/51 \\ 4/85 & 83/255 & 32/51 \\ 58/85 & 1/51 & 76/255 \end{pmatrix}, \quad (16)$$

$$E_- = \begin{pmatrix} 10/17 & -5/51 & -25/51 \\ 8/17 & -4/51 & -20/51 \\ -10/17 & 5/51 & 25/51 \end{pmatrix}. \quad (17)$$

For $t \neq 1$, put

$$S_t = \begin{pmatrix} 1-t & t & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad K_t = S_t^{-1} K S_t. \quad (18)$$

Define $q(t) = 35 - 38t - 12t^2$. Exact multiplication gives the audited formula

$$K_t = \begin{pmatrix} \frac{15-4t}{85} & \frac{q(t)}{255(1-t)} & \frac{35-32t}{51(1-t)} \\ \frac{4(1-t)}{85} & \frac{83+12t}{255} & \frac{32}{51} \\ \frac{58(1-t)}{85} & \frac{5+174t}{255} & \frac{76}{255} \end{pmatrix}. \quad (19)$$

Theorem 4.1 (Positive row-stochastic isospectral interval). *For every*

$$\boxed{-\frac{5}{174} < t < \frac{1}{2}}, \quad (20)$$

the matrix K_t is strictly positive and row-stochastic. For every such t ,

$$\text{spec}(K_t) = \{1, -2/5, 1/5\}, \quad (21)$$

and for every integer $m \geq 1$,

$$\boxed{\text{Tr } K_t^m = 1 + (-2/5)^m + (1/5)^m.} \quad (22)$$

Proof. On (20), every denominator in (19) is positive. The nonconstant numerators

$$15 - 4t, \quad q(t), \quad 35 - 32t, \quad 83 + 12t, \quad 5 + 174t$$

are also positive; for q , it suffices that $t < 1/2$ because its positive root exceeds $1/2$. More directly, $q'(t) = -38 - 24t < 0$ on the stated interval and $q(1/2) = 13$, so $q(t) > 13$. Thus every entry is positive. Since $S_t \mathbf{1} = \mathbf{1}$, also $S_t^{-1} \mathbf{1} = \mathbf{1}$, and $K_t \mathbf{1} = S_t^{-1} K \mathbf{1} = \mathbf{1}$.

Similarity preserves the spectrum and gives $K_t^m = S_t^{-1} K^m S_t$. RH-335 supplies the three distinct eigenvalues of K , so trace invariance proves (22). $\square$

Remark 4.2 (Sufficient versus maximal). The interval (20) is convenient, not maximal. Intersecting all strict entry inequalities in the connected component containing zero gives

$$\left(-\frac{5}{174}, \frac{-19 + \sqrt{781}}{12} \right). \quad (23)$$

The upper endpoint is the positive root of $q(t)$. No maximality claim is made beyond this explicit finite family.

5 Moving projector masses and corrected singleton cells

Let

$$E_-(t) = S_t^{-1} E_- S_t. \quad (24)$$

It is the Riesz projector of K_t at $-2/5$ by similarity.

Proposition 5.1 (Exact projector-mass drift). *For the three singleton windows,*

$$\boxed{\pi(t) = \text{diag } E_-(t) = \left(\frac{10 - 8t}{17}, \frac{-4 + 24t}{51}, \frac{25}{51} \right).} \quad (25)$$

Therefore

$$\pi(t) - \pi(0) = \frac{8t}{17}(-1, 1, 0), \quad \sum_i \pi_i(t) = 1. \quad (26)$$

Proof. Insert (17) and (18) into (24) and read the diagonal. Subtraction gives (26); its entries sum to zero. $\square$

Use now the RH-335 fixed-order fixture $n = 2$, $r_H = 17/20$, parity eigenvalue $-2/5$, and zero deterministic singleton slots. Define

$$C_i(t) = \left(\frac{20}{17} \right)^2 \left\{ (K_t^2)_{ii} + \frac{21}{25} \pi_i(t) \right\}. \quad (27)$$

Theorem 5.2 (Isospectral corrected-cell obstruction). *The corrected singleton cells are*

$$\mathcal{C}(t) = \left(\frac{6800 - 5760t}{4913}, \frac{400 + 5760t}{4913}, \frac{6672}{4913} \right). \quad (28)$$

For every admissible t ,

$$\sum_i \mathcal{C}_i(t) = \frac{48}{17}, \quad (29)$$

$$\mathcal{C}(t) - \mathcal{C}(0) = \frac{5760t}{4913}(-1, 1, 0). \quad (30)$$

At $t = 1/100$, the drift is exactly

$$\left(-\frac{288}{24565}, \frac{288}{24565}, 0 \right). \quad (31)$$

Proof. Similarity applied to K^2 gives

$$\text{diag } K_t^2 = \left(\frac{215 - 192t}{425}, \frac{53 + 192t}{425}, \frac{242}{425} \right).$$

Combine this with (25) in (27) and simplify to (28). Summation and subtraction prove (29)–(30); setting $t = 1/100$ gives (31). $\square$

The full spectrum and all power traces are constant, yet the corrected singleton cells move. This proves nonidentification of those cells from global spectral data in a strictly positive row-stochastic family. RH-210 already proved the general logical phenomenon that projectors may move under a fixed divisor [4]. The narrow addition here is the simultaneous positivity, Markov normalization, all-power trace lock, and explicit corrected-cell ledger.

The family remains nonphysical finite algebra. In particular, fixed $n = 2$ is not a $k = 1$ first-alias counterloop: the archived counterloop definition requires $k \geq 2$.

6 Reproduction and claim boundary

The artifact evaluates every matrix identity with exact rational arithmetic. It checks the displayed formulas, strict positivity at rational sample points inside the proved interval, the endpoint factorization, two-sided projector intertwining, twelve power traces, projector masses, corrected cells, invariant sums, and the exact one-percent drift. Floating-point rows only reproduce the symbolic phase conversion and the diagnostic exponent; they are not interval certificates.

RH-336 proves the moving projector-mass threshold and a nonphysical isospectral corrected-cell obstruction. It does not identify a physical $\Delta_B + \Delta_S$ cancellation, prove a physical nonzero normalized obstruction, exclude raw/local/alias signed cancellation, close the far or off-alias terms, transport the noisy head, or glue a determinant. No result constructs a Hilbert–Polya operator, identifies Riemann zeros, proves a von Mangoldt trace formula or completed-zeta divisor equality, or proves the Riemann hypothesis. Gates A–E remain false/open.

References

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- [3] Bin Wang. Projector-localized parity ledgers and cellwise extension nonuniqueness, 2026. Repository paper RH-335.
- [4] Liang Wang. A divisor-first pivot inside gate a, 2026. Repository paper RH-210.